Is a Description Worth a Thousand Demonstrations? Measuring the Effective Sample Size of Task Descriptions in In-Context Learning

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Abstract

Prompts for in-context learning usually pair demonstrations with a task description, yet theories of in-context learning mostly model the demonstrations alone. We ask how many demonstrations a description is worth: its *effective sample size* (ESS), the number of demonstrations that lower a learner's error as much as the description does. For the Bayes-optimal predictor in in-context linear regression, the ESS depends on the description's *specificity* (how much uncertainty about the task it removes) and its *reliability* (the probability that it is relevant). We prove that a reliable description's ESS grows linearly with the task dimension when the description is loose and with its precision when it is specific. Without demonstrations, a description that may be irrelevant is worth a bounded number of demonstrations however specific it is; with them, trusting it fully can be worse than ignoring it. Meta-trained Transformers come close to the Bayes-optimal ESS for reliable descriptions and take their trust from training. Pretrained language models answering without reasoning adjust to neither the stated specificity nor the stated reliability as the Bayes-optimal predictor does: a description worth 36 demonstrations to it is worth 1 to 8 to them (medians over one to four demonstrations shown). Two models that reason first come close to the Bayes-optimal weight on a relevant description.

1. Introduction

Prompts often tell a model what the task is before, or instead of, showing it. A few-shot prompt for in-context learning (ICL) pairs a natural-language task description with demonstrations (Brown et al., 2020), and deployed models also receive descriptions as system prompts, task instructions, and the instruction files that configure coding agents (Chatlatanagulchai et al., 2025). Yet we know little about how a model uses such a description or how effective it is in different forms, and the evidence is mixed. A prompt can be worth hundreds of data points in prompt-based fine-tuning (Le Scao & Rush, 2021), and a written summary can stand in for many demonstrations (Honda et al., 2025). Models, however, learn as fast from irrelevant or misleading instructions as from instructive ones (Webson & Pavlick, 2022), let in-context evidence outweigh explicit instructions (Gupta et al., 2025), and coding agents given repository instruction files do not generally solve more tasks (Gloaguen et al., 2026).

Theories of ICL, in contrast, mostly model the prompt as a sequence of demonstrations (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023); those that add a description show that a learner can use it to predict better (Huang & Ge, 2025), but neither put a value on it nor say what it is worth when it may be wrong. We measure the value of a task description by the number of demonstrations it can replace, and ask what determines that number.

We take the Bayesian view of ICL (Xie et al., 2022): the prompt is evidence about a latent task, a demonstration is one noisy observation of it, and a description is a statement about which task generated the demonstrations, so it sharpens the prior; we set aside instruction following. Bayesian statistics measures the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information (Morita et al., 2008). We define the ESS of a description in the same way (Figure 1) and compute it in noisy in-context linear regression with a Gaussian task prior (Garg et al., 2022; Akyürek et al., 2023; Raventós et al., 2023), where Bayes-optimal prediction has a closed form, and then measure it for meta-trained Transformers and for pretrained language models given a stated prior.

Following the Bayesian literature on informative priors, we characterize a description by two properties. The first is its *specificity*, the precision of the prior it gives (Morita et al., 2008; Neuenschwander et al., 2020). "Write this report like an accountant" only loosely indicates the task; a descrip-

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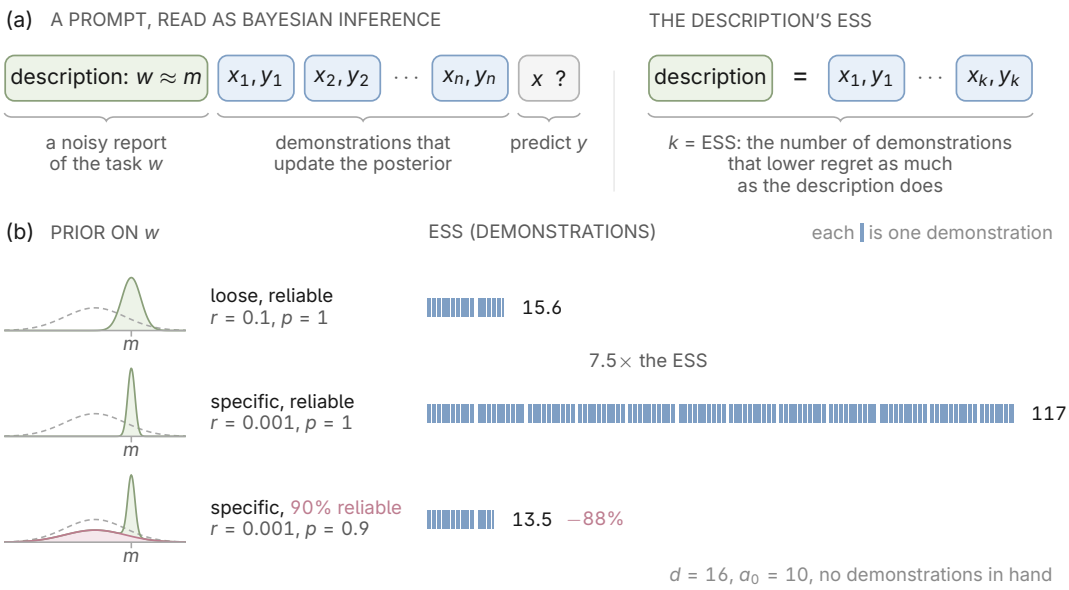


Figure 1. **A description is worth a number of demonstrations, set by its specificity and its reliability.** (a) A prompt read as Bayesian inference: a description states that the task w is near a value m, and the demonstrations update the posterior. The ESS of the description is the number of demonstrations it replaces. (b) The prior on w without the description (dashed) and given three descriptions (solid; sketched, not to scale), and the ESS of each with no demonstrations in hand, at task dimension $d = 16$ and prior signal-to-noise ratio $a_0 = 10$; r is the fraction of the prior variance a relevant description leaves, p the probability that it is relevant, and pink marks the component for an irrelevant description. Specificity raises the ESS (§4); a one-in-ten chance of being irrelevant removes most of that gain (§5).

tion that fixes the report's sections, figures, and accounting standard all but states it. The second is its *reliability*, the probability that it is relevant, which sets the weight of that prior in a robust mixture that guards against conflict with the data (Schmidli et al., 2014; Evans & Moshonov, 2006): instructions written for one report can be pasted into a prompt for another, or simply be wrong. Our contributions are:

- **A common unit.** We define the ESS of a task description, the number of demonstrations that lower a learner's error as much as the description does (Definition 1). It applies to any learner, and with squared error it needs only the learner's answers.
- **Theory.** For the Bayes-optimal predictor, a reliable description's ESS grows linearly in the task dimension when the description is loose and in its precision when it is specific (Proposition 1). With no demonstrations in hand, one that may be irrelevant is worth a bounded number however specific it is, and once demonstrations are in hand, trusting it fully can be worse than ignoring it (Proposition 2, Corollary 3, Appendix A.3).
- **Experiments.** Meta-trained Transformers come close to the Bayes-optimal ESS for reliable descriptions (§6). Pretrained language models answering without reasoning adjust to neither a description's stated specificity nor its stated reliability as the Bayes-optimal predictor does: a description worth 36 demonstrations to it is worth a median of 1 to 8 to them. Two models that reason first come close to the Bayes-optimal weight on a relevant description (§7).

2. Related Work

ICL as Bayesian inference. Xie et al. (2022) cast ICL as implicit Bayesian inference. Transformers trained on samples from a prior approximate its posterior predictive (Müller et al., 2022), and meta-trained Transformers have been shown to implement Bayes-optimal estimators such as ridge regression under a Gaussian task prior, given enough task diversity (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024).

Task descriptions in theories of ICL. Some theories of ICL include task descriptions, as tokens that indicate a hypothesis class (Lin et al., 2025) or carry the input mean (Huang & Ge, 2025), or as instructions in a generalization bound (Tong et al., 2026). They generally do not value a description in demonstrations or model the chance that it is irrelevant. Closest to our setting, Zhu et al. (2026) show that a Transformer given a prefix of related datasets performs amortized hierarchical Bayesian prediction; we express the value of such conditioning information in demonstrations and allow it to be irrelevant.

Effective sample size of a prior. Our ESS follows Bayesian methods for the sample size of a prior (Morita

et al., 2008; Neuenschwander et al., 2020; Reimherr et al., 2021). As in Reimherr et al. (2021) and the effective current sample size of Wiesenfarth & Calderazzo (2020), it depends on the number of observations in hand, and as in Reznik (2026), who sets a power prior's discount on possibly mismatched historical data, it measures information by predictive loss. With many observations in hand it returns to the ESS of Morita et al. (2008) (§4). To model a description that may be irrelevant we use a robust mixture prior, which clinical trials use to discount historical data that may conflict with the current trial (Schmidli et al., 2014).

Descriptions and priors in pretrained language models. Cook et al. (2026) value instructions in units of training examples and find that one instruction can stand in for up to 100 execution examples in fine-tuning data. Wenliang et al. (2026) treat a prompt as conditioning for a Bayesian sequence predictor and find that an optimized prompt can be shorter than demonstrations of equal effect. Tests of pretrained models against a known optimum on synthetic tasks find that in-context evidence outweighs a stated prior (Gupta et al., 2025), that learning curves approach a Bayesian reference (Akata et al., 2026), and that models estimate a density from numbers in context like a kernel estimator of adaptive width (Liu et al., 2025). Arora et al. (2025) model in-context learning curves with a Bayesian scaling law whose terms include a task prior; text given alongside numbers also shapes a model's numerical predictions (Requeima et al., 2024). We add a description with a stated specificity and reliability to such a test and measure its ESS against the Bayes-optimal ESS. Pretrained models can do well with wrong demonstration labels (Min et al., 2022), and larger ones override their priors with in-context labels more readily (Wei et al., 2023); we ask the converse, whether a model weighs a description by what the prompt says about it.

3. Problem Setup

Tasks and demonstrations. A task is a weight vector $w \in \mathbb{R}^d$. A demonstration is an input $x \sim \mathcal{N}(0, I_d)$ with a label $y \sim \mathcal{N}(w^\top x, 1)$; we measure labels in units of their noise. The baseline prior over tasks is $w \sim \mathcal{N}(0, a_0 I_d)$, so a_0 is the prior signal-to-noise ratio. The numerical results of §§4–6 use $a_0 = 10$.

Descriptions. A description is a vector $m \in \mathbb{R}^d$, governed by two parameters, r and p, that we interpret below. A latent indicator $Z \sim \text{Bernoulli}(p)$, independent of m, says whether the description is *relevant* ($Z = 1$) or *irrelevant* ($Z = 0$):

$$
\begin{aligned}
m &\sim \mathcal{N}(0, (1-r)\, a_0 I_d) \\
w \mid m, Z = 1 &\sim \mathcal{N}(m, r a_0 I_d) \\
w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d)
\end{aligned}
$$

In words, a relevant description says the task lies near m, within a variance ra_0 in every direction; an irrelevant one is unrelated to the task. Either way, w marginally has the baseline prior. Marginalizing over Z, the prior for w given m is the robust mixture $p\,\mathcal{N}(m, ra_0 I_d) + (1-p)\,\mathcal{N}(0, a_0 I_d)$ (Schmidli et al., 2014). The parameter $r \in (0, 1)$ is the fraction of the prior variance that a relevant description leaves, and it sets the description's **specificity**: the smaller r, the more specific the description, from one that only loosely indicates the task (r near 1) to one that all but states it (r near 0). The parameter p is the description's **reliability**. We call a description with $p = 1$ *reliable*, and write $c = 1/(ra_0)$ for the precision of a relevant description's prior, in units of demonstrations. We call a description *loose* when c is small next to d and *specific* when c exceeds d; at $d = 16$ and $a_0 = 10$, $r = 0.1$ ($c = 1$) is loose and $r = 0.001$ ($c = 100$) is specific.

Regret and ESS. A predictor sees a prompt D (the description, n demonstrations, and a query x) and predicts the query's label. As Garg et al. (2022) do, we score its prediction $\hat{y}$ by its error, the squared distance from the noiseless label $w^\top x$:

$$
R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big], \tag{1}
$$

the expected squared error less the noise variance. Let $R^{\text{desc}}(n)$ be a predictor's error with the description and n demonstrations, and $R^{\text{demo}}(n)$ its error with n demonstrations alone. In §§4–5 the predictor is Bayes-optimal: it predicts the posterior mean of $w^\top x$, so its error is the expected posterior variance of $w^\top x$, and $R^{\text{desc}}_{p=1}(n)$ is the value of $R^{\text{desc}}(n)$ for a reliable description. The ESS counts the further demonstrations that a predictor without the description would need to match the predictor with it.

Definition 1. Given n demonstrations, the effective sample size (ESS) of a description is $\text{ESS}_n = n^* - n$, where n^* is the smallest $t \geq 0$ with $R^{\text{demo}}(t) = R^{\text{desc}}(n)$, with R^{demo} interpolated linearly between the values of t at which it is known. If $R^{\text{demo}}(t) < R^{\text{desc}}(n)$ for every t, we write $\text{ESS}_n \leq -n$: the description does more harm than all n demonstrations do good. If $R^{\text{desc}}(n)$ is below R^{demo} at the largest known t, we write $\text{ESS}_n > t - n$.

The errors average over tasks, descriptions, and demonstrations, so the ESS belongs to a kind of description, set by r and p, not to one value of m. For the Bayes-optimal predictor the ESS is never negative: conditioning on the description cannot raise the expected posterior variance, so $R^{\text{desc}}(n) \leq R^{\text{demo}}(n)$, and R^{demo} decreases in n (Appendix A.1). For any other learner the ESS is read from that learner's own two error curves (§§6–7), so it measures what the description is worth to that learner, and it is negative when the description hurts.

4. Specificity: Reliable Descriptions

A reliable description narrows the prior over the task; Proposition 1 counts the demonstrations that would narrow it as much. Write $W_n = X_n^\top X_n$, with the n demonstration inputs as the rows of X_n. Without the description the posterior covariance of w is $(I/a_0 + W_n)^{-1}$, and a reliable description replaces the baseline prior's precision $1/a_0$ with its own, c. Averaged over the query, the error is the trace of the posterior covariance:

$$R^{\text{demo}}(n) = \mathbb{E}\,\text{tr}(I/a_0 + W_n)^{-1},$$
$$R^{\text{desc}}_{p=1}(n) = \mathbb{E}\,\text{tr}(cI + W_n)^{-1}.$$

Proposition 1 (ESS of a reliable description)**.** *Let $p = 1$.*

(i) (Loose descriptions, worth about $(1-r)d$) $\text{ESS}_0 \geq (1-r)\,d$*, and* $\text{ESS}_0 \to (1-r)\,d$ *as $a_0 \to \infty$ with r fixed.*

(ii) (Specific descriptions, worth $c+d+1$) If c is a whole number, $\text{ESS}_0 \to c+d+1$ *as $a_0 \to \infty$ with c fixed.*

Appendix A.1 proves it from two standard facts. With $n < d$ demonstrations the data say nothing about $d-n$ directions, so $R^{\text{demo}}(n) \geq (d-n)\,a_0$. With a flat baseline prior and $n \geq d+2$, $R^{\text{demo}}(n) = d/(n-d-1)$, the trace of the mean of an inverse Wishart matrix (Muirhead, 1982). A reliable description leaves $R^{\text{desc}}_{p=1}(0) = d/c = ra_0d$, and the ESS is where these curves fall to that value.

The ESS has two parts. The first, the *dimension term*, counts dimensions: each of the first d demonstrations removes the uncertainty in one direction. A loose description removes a fraction $1-r$ of the prior variance in every direction and is worth about $(1-r)d$ of them; a specific one removes nearly all of it and is worth about d. A specific description is also worth c, the precision of its prior in demonstrations. This is the ESS that Morita et al. (2008) give a normal prior, the ratio of the observation variance to the prior variance, here $1/(ra_0) = c$. With many demonstrations in hand the dimension term fades, and the ESS returns to this value, less the precision $1/a_0$ that the baseline prior already supplies: $\text{ESS}_n \to (1-r)\,c$ (Corollary 3(ii)).

At $d = 16$ and $a_0 = 10$, a description that leaves a tenth of the prior variance ($r = 0.1$, $c = 1$) is worth 15.6 demonstrations, against $(1-r)d = 14.4$, and one that leaves a thousandth ($r = 0.001$, $c = 100$) is worth 117, matching $c+d+1$ (Figure 2). With many demonstrations the dimension term fades, because they have already removed that uncertainty: at $r = 0.1$, 100 demonstrations in hand lower the ESS from 15.6 to 1.1.

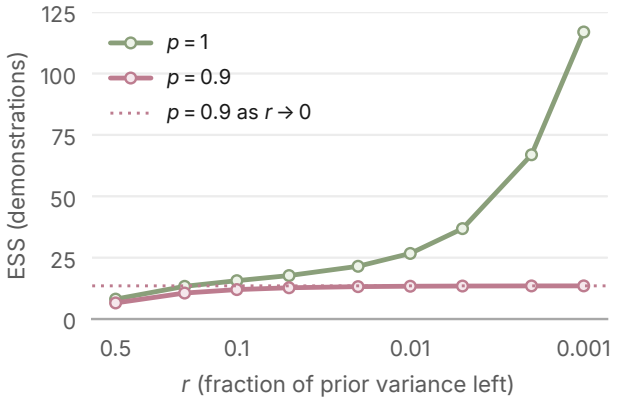


Figure 2. **Specificity raises the ESS of a reliable description without bound; one that may be irrelevant stays below a bound.** Bayes-optimal ESS of a description with no demonstrations in hand, against r, the fraction of the prior variance it leaves, for a reliable description ($p = 1$) and one that is relevant with probability 0.9. Dotted: the limit of the $p = 0.9$ ESS as $r \to 0$. $d = 16$, $a_0 = 10$; eight batches of 4,000 prompts, standard errors below 0.02 demonstrations.

5. Reliability: Descriptions That May Be Irrelevant

Proposition 2 gives the price of not knowing whether a description is relevant, and the error in closed form when no demonstrations are in hand. When $p < 1$ the Bayes-optimal prediction mixes two predictions, $\mu_1^\top x$, which takes the description as relevant, and $\mu_0^\top x$, which ignores it (μ_1 and μ_0 are the posterior means of w under the two readings), weighted by the posterior probability π_n that the description is relevant.

Proposition 2 (Error of a description that may be irrelevant)**.** *For every $n \geq 0$,*

$$R^{\text{desc}}(n) \;=\; p\,R^{\text{desc}}_{p=1}(n) + (1-p)\,R^{\text{demo}}(n) + \mathcal{V}_n, \quad (2)$$

where $\mathcal{V}_n = \mathbb{E}\big[\pi_n(1-\pi_n)\big((\mu_1-\mu_0)^\top x\big)^2\big] \geq 0$. With no demonstrations in hand,

$$R^{\text{desc}}(0) = \big(1 - p^2(1-r)\big)\,a_0 d. \quad (3)$$

Appendix A.2 gives the proof. The first two terms are the error of a *cheating* predictor, one that is told whether the description is relevant: with probability p it holds a reliable description, and with probability $1-p$ it holds only the demonstrations. To that predictor, a description that may be irrelevant is worth what a reliable one is worth when it is relevant and nothing when it is not.

The last term, $\mathcal{V}_n$, is the price of not being told: the variance of the prediction between the two readings of the description. It is small when the readings agree or when the demonstrations make one reading far likelier than the other:

with no demonstrations it grows as a more specific description pulls the readings apart, but with demonstrations a specific description is also easier to tell from an irrelevant one. With no demonstrations, (3) puts the two predictors side by side: of the error $a_0 d$ that the baseline prior leaves, the cheating predictor removes a fraction $p(1-r)$ and the Bayes-optimal predictor $p^2(1-r)$, so not being told costs $p(1-p)(1-r)\,a_0 d$.

Corollary 3 (ESS of a description that may be irrelevant)**.**

(i) (Without demonstrations) If $p < 1$, then $\mathrm{ESS}_0 \le n_p^{\max}$, where $n_p^{\max}$, which depends on d, a_0, and p but not on r, solves $R^{\mathrm{demo}}(n_p^{\max}) = (1-p)\,R^{\mathrm{demo}}(0)$ for the interpolated R^{demo}. Also $\mathrm{ESS}_0 \ge p^2(1-r)\,d$, and as $a_0 \to \infty$ with r fixed, $n_p^{\max} \to p\,d$ and $\mathrm{ESS}_0 \to p^2(1-r)\,d$.

(ii) (With many demonstrations) As $n \to \infty$, $\mathrm{ESS}_n \to (1-r)\,c$ if $p = 1$, and if $p < 1$,

$$\limsup_{n\to\infty} \mathrm{ESS}_n \le p\,(1-r)\,c.$$

Without demonstrations the ESS is bounded. Refining a description from $r = 0.1$ to $r = 0.001$ raises its ESS from 15.6 to 117 if it is reliable, but only from 12.0 to 13.5 if it is relevant with probability 0.9. However specific the description, the average error cannot fall below the share contributed by irrelevant descriptions, $(1-p)R^{\mathrm{demo}}(0)$, and Corollary 3(i) turns that floor into a bound on the ESS: $n_p^{\max} = 15.6$ demonstrations at $d = 16$, $a_0 = 10$, and $p = 0.9$, the ESS of a reliable description that leaves a fraction $1-p$ of the prior variance, which leaves the same error. The bound holds even for a predictor told whether the description is relevant, so it comes from irrelevant descriptions being worth nothing, not from uncertainty about relevance. That uncertainty lowers the ESS further, from about pd to about $p^2 d$ at a high signal-to-noise ratio: as $r \to 0$ the ESS approaches 13.5 (dotted in Figure 2), against $p^2 d = 13.0$.

With many demonstrations, the ESS is at most p times a reliable description's. Demonstrations shrink the error in the irrelevant case, $R^{\mathrm{demo}}(n)$, so the floor $(1-p)R^{\mathrm{demo}}(n)$ falls and the ESS can grow. In the limit a reliable description is worth $(1-r)\,c$ demonstrations: the dimension term is gone, and what remains is the precision the description adds to the baseline prior. A description that may be irrelevant is then worth at most p times as much. With 100 demonstrations in hand, the $r = 0.001$ description is worth 109 demonstrations if reliable and 86.6 at $p = 0.9$, against limits of 99.9 and at most 89.9; the reliable value exceeds its limit because at $n = 100$ it still carries part of the dimension term.

Full trust can be worse than no description. Unlike the Bayes-optimal predictor, which hedges, a *fully trusting* predictor treats every description as relevant, predicting as if $p = 1$. With no demonstrations its error is

$$R^{\mathrm{trust}}(0) = \big(1 - (2p-1)(1-r)\big)\,a_0 d \tag{4}$$

(Appendix A.3): it removes a fraction $(2p-1)(1-r)$ of the prior error, against $p^2(1-r)$ for the Bayes-optimal predictor, and does worse than ignoring the description only when $p < 1/2$. Demonstrations do not rescue it. They show whether a specific description is relevant, but the trusting predictor keeps predicting near the stated m: as $r \to 0$ its error tends to $2(1-p)\,a_0 d$ at every n, which demonstrations alone eventually undercut. At $d = 16$, $a_0 = 10$, and $p = 0.9$, the fully trusting predictor with the $r = 0.001$ description is worse than no description from 15 demonstrations on; with 20 in hand its error is 23.8, against 4.3 without the description.

6. Meta-Trained Transformers

The Bayes-optimal predictor knows how specific and how reliable descriptions are; a network must learn both from its training distribution. We meta-train Transformers on the setting of §3 and ask how close they come to the Bayes-optimal ESS and what sets their trust in a description.

Setup. Prompts use the prefix embedding of Huang & Ge (2025): an optional description token carrying m, up to 20 demonstration tokens, and a query token. The network has the size of Garg et al. (2022)'s (12 layers, 8 heads, width 256) and their optimizer, but is a bidirectional Transformer without positional encoding (Appendix B). To keep training short we use $d = 5$, with $a_0 = 10$ as before. The network outputs a mean and a log variance for the query's label and is trained with log loss, as in Zhu et al. (2026). Because it outputs a distribution, we score it, and the Bayes-optimal predictor it is compared with, by log-loss regret against the oracle, $\mathbb{E}[-\log f(y \mid D) + \log \phi(y; w^\top x, 1)]$ with ϕ the Gaussian density (Genewein et al., 2025), which judges the predicted variance as well as the mean. The description token carries only m, and half the training prompts omit it, so each network must learn r and p from its training distribution. We train six networks, one for each specificity ($r = 0.1$, loose; $r = 0.01$, specific) and reliability ($p \in \{1, 0.99, 0.9\}$), plus a second seed at $r = 0.01$, $p = 0.9$ and a network at $r = 0.05$, $p = 0.9$ for a longer-training check.

Evaluation. We test each network on 20,000 fresh prompts at every n from 0 to 20 (0 to 15 for the $r = 0.05$ network), with and without a description, at its training reliability and at $p = 1$ and 0.9 (the second seed also at 0.99; Table 1). Its ESS follows Definition 1 with its own regret curves; without a description, every network's regret is within 0.004 nats of the Bayes-optimal predictor's at $n = 0$ and within 0.03 nats on average over $n = 0$ to

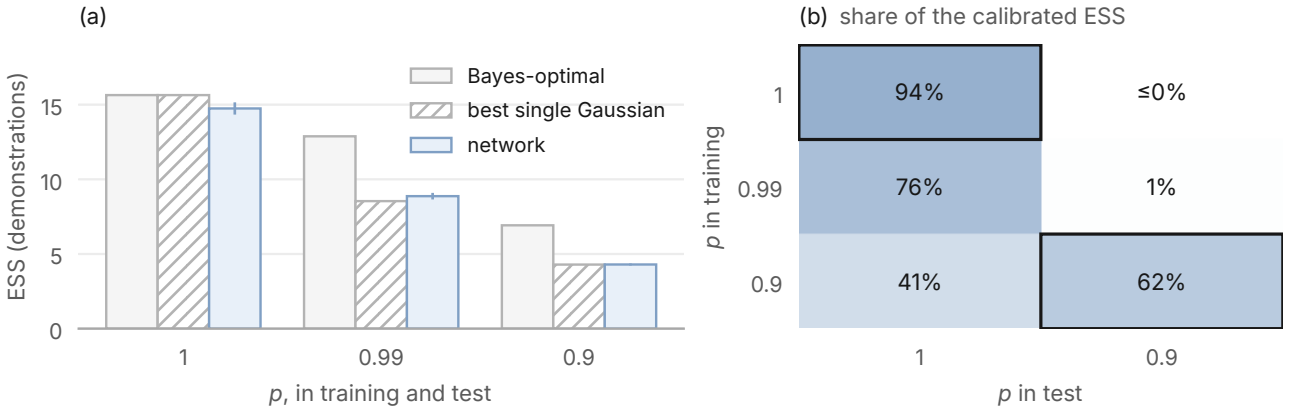


Figure 3. **Networks come close to the Bayes-optimal ESS as far as one Gaussian output can, and take their trust from training.** Specific description ($r = 0.01$), $d = 5$, $a_0 = 10$, no demonstrations; error bars, $\pm$ one standard error over 20,000 prompts. (a) Trained and tested at the same reliability p: the Bayes-optimal ESS, the best a single Gaussian output can reach (at $p = 1$ the Bayes-optimal predictive is itself Gaussian), and the network; the network's ESS is read from its own no-description curve, the others' from the Bayes-optimal one, so the network can exceed the single-Gaussian bar. (b) Trained at one reliability and tested at another: the network's ESS as a share of the calibrated predictor's; outlined, training matches test. Table 1 (Appendix B) gives every value, including those for the loose description ($r = 0.1$).

20. We compare the network's ESS with the *trained* Bayes-optimal predictor, which uses the reliability the network was trained on, the *calibrated* one, which uses that of the test prompts, and the *best single Gaussian*, which has the mean and variance of the Bayes-optimal predictive. A network's *implied trust*, a summary of its mean predictions only, is the p on a grid whose Bayes-optimal mean predictions with no demonstrations are closest to the network's (Table 1).

Networks come close to the Bayes-optimal ESS as far as their output allows. Trained and tested on reliable descriptions, the networks come close to the Bayes-optimal ESS: 14.7 ± 0.4 ($\pm$ one standard error; 13.2 if read from the Bayes-optimal no-description curve) against 15.6 for the specific description (Figure 3a), and 5.24 ± 0.05 against 5.25 for the loose one (Table 1). With descriptions that may be irrelevant and no demonstrations, they fall short: 4.30 ± 0.07 against 6.92 at $p = 0.9$, $r = 0.01$ (a second seed gives 4.26 ± 0.08). The output accounts for the shortfall. The Bayes-optimal predictive is then a mixture of two Gaussians, which a network that emits one mean and one variance cannot express; the networks match the best single Gaussian instead (4.29 in that case). As demonstrations accumulate, the posteriors under the two components concentrate on the same task and their predictives converge, so one Gaussian comes closer to the mixture, and the networks close part of the gap: with five demonstrations in hand that network's ESS is 7.3, against 8.6 for the Bayes-optimal predictor and 8.4 for the best single Gaussian. The $r = 0.05$ network keeps its shortfall after twice the training steps (3.9 against 5.0; Appendix B).

Training sets a network's trust. With no demonstrations the prompt says nothing about relevance (Appendix A.2, Step 1), so any learner must take its trust from training; Figure 3b shows how well each network hedges on prompts whose reliability differs from its training. The network trained on specific, reliable descriptions treats every description as relevant (implied trust 0.999). When one test description in ten is irrelevant, the description hurts: the network's regret is 2.87 nats with it and 1.87 without, so its ESS is at most zero, against 6.9 for the calibrated predictor. Tested at $p = 0.9$, the network trained at $p = 0.99$ also keeps almost none of the description's value (ESS 0.1 ± 0.4), where the trained predictor, a mixture that also assumes $p = 0.99$, keeps 6.1; one Gaussian fitted to rare irrelevance cannot express that mixture. Networks trained at $p = 0.9$ err the other way. They keep hedging, at an implied trust of 0.85 to 0.9 across the two seeds, even when every test description is relevant, and reach an ESS of 6.4 and 6.6 where the calibrated predictor reaches 15.6 and the trained one 11.9.

7. Pretrained Language Models

The networks learn a description's specificity and reliability from training; a pretrained language model can learn them only from the prompt. We therefore state both in the prompt, together with the noise level, and compare the description's ESS for the model with its ESS for a predictor that is Bayes-optimal given what the prompt states. The design follows Gupta et al. (2025) and Akata et al. (2026): a stated prior is a description whose Bayes-optimal ESS is known exactly, so any shortfall is the model's.

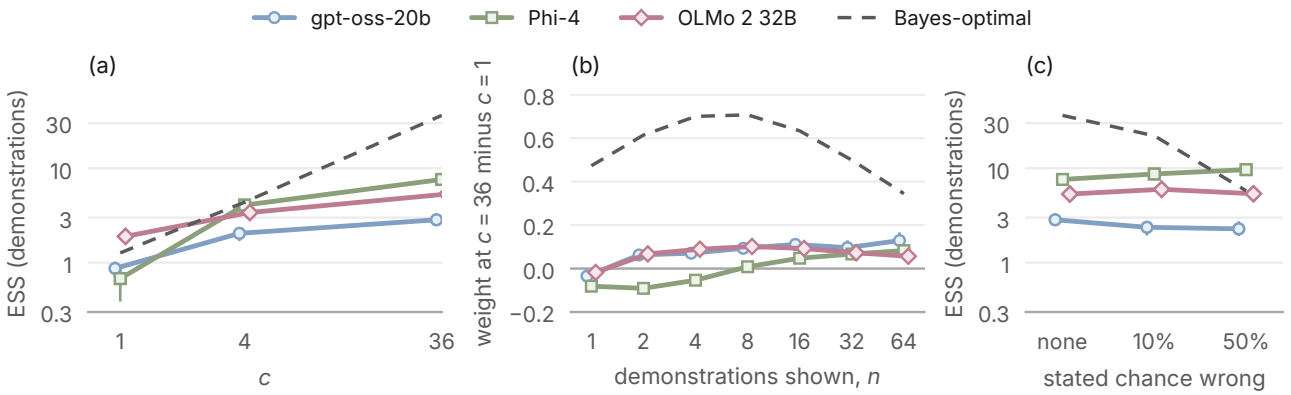


Figure 4. **Language models barely adjust to a description's stated specificity or reliability.** gpt-oss-20b, Phi-4, and OLMo 2 32B, read from their first-token distribution and answering without reasoning (the other three are in Table 2, Appendix C), against the Bayes-optimal predictor (dashed), for a relevant description; $\sigma = 60$, 2,000 tasks; error bars, 95% intervals over tasks (bootstrap in (a) and (c), normal approximation in (b)); in (a) and (c) the ESS is the median over $n = 1$, 2, and 4 demonstrations shown. (a) ESS of the descriptions with Bayes-optimal ESS $c = 1$, 4, and 36. (b) Weight on the stated value for $c = 36$ minus that for $c = 1$, against the number of demonstrations shown. (c) ESS of the $c = 36$ description with no stated chance of being wrong, or a stated 10% or 50% chance; the description is in fact relevant, and the Bayes-optimal predictor takes the stated chance at face value.

Setting. We use the simplest case of §3: one weight ($d = 1$) and a fixed input $x = 1$. A task is then a hidden mean $w \sim \mathcal{N}(500, 100^2)$, and a demonstration is a number $y \sim \mathcal{N}(w, \sigma^2)$, rounded to an integer. A description states a value m and a standard deviation τ, and tasks are drawn so that $w \mid m \sim \mathcal{N}(m, \tau^2)$: the description of §3 with $r = \tau^2/100^2$. We set the noise to $\sigma = 60$ and τ to 60, 30, or 10, and show $n = 0$, 1, 2, 4, 8, 16, 32, or 64 demonstrations; a second noise level, $\sigma = 30$ with $\tau = 30$, 15, or 5, serves as a check. An *irrelevant* description states the value that a relevant description of an independently drawn task would state, so it is unrelated to the task but not necessarily far from it. A *stated reliability* adds one sentence giving a 10% or 50% chance that the description is wrong: the mixture of §3 with $p = 0.9$ or 0.5, in words. The description is one sentence of the prompt, such as "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30" (Appendix C shows a full prompt).

Bayes-optimal predictor. The Bayes-optimal predictor here knows only what the prompt says: the noise σ and the description. Beyond that the mean is unknown, which we take as uniform over the three-digit numbers (Appendix C gives the ESS under the prior the tasks are drawn from); the predictor is optimal for what the prompt states, not for how the tasks are drawn, and we call it Bayes-optimal in that sense. Given a relevant description and n demonstrations whose mean is $\bar{y}$, it predicts $(c\,m + n\,\bar{y})/(c + n)$ with $c = \sigma^2/\tau^2$, a weight of $c/(c + n)$ on the stated value. The description adds precision $1/\tau^2 = c/\sigma^2$, that of c demonstrations. The squared error is therefore $\sigma^2/(c + n)$ with it and σ^2/n without, and the description's Bayes-optimal ESS is c at every n, up to the truncation of that uniform prior. The predictor takes an irrelevant description as relevant unless told otherwise, so an irrelevant description has a negative Bayes-optimal ESS here, and unlike in §§4–6 a model can do better than this predictor. Our values of τ give $c = 1$, 4, and 36 at both noise levels, and we name the descriptions by c.

Models. We use six open chat models. gpt-oss-20b (OpenAI, 2025), Phi-4 (14B; Abdin et al., 2024), and OLMo 2 32B Instruct (Team OLMo et al., 2025) encode every three-digit integer as one token, so the reply's first-token distribution, restricted to the 900 number tokens and renormalized, is the model's distribution over the next number, and its prediction is that distribution's mean. Qwen3.5 9B (Qwen Team, 2026), Gemma 4 12B, and Gemma 4 31B (Gemma Team, 2026) split numbers into digits, so their prediction is the number they write under greedy decoding; on gpt-oss-20b the two readouts give weights within 0.2 of each other (Appendix C). Three others begin fewer than three quarters of their replies with a number and are not reported. The prompt is the user turn of the chat template (Appendix C).

Measurement. A model's error is the root-mean-square distance from the hidden mean over 2,000 tasks. We apply Definition 1 with the squared error of §3, and read the ESS off the model's own no-description error curve, interpolated in $\log n$ in place of linearly. We summarize the ESS by its median over $n = 1$, 2, and 4, with 95% bootstrap intervals over tasks (Table 2; every n in Table 8 and Appendix C). Computed the same way, the Bayes-optimal ESS is 1.3, 4.4, and 36 rather than 1, 4, and 36, because the interpolation and the finite sample of tasks shift it (Appendix C). A model's *weight* on the stated value is the coefficient on m in

a regression of its prediction on m and the demonstrations' mean $\bar{y}$ across tasks, with a constant; it is $c/(c+n)$ for the Bayes-optimal predictor.

A specific description is worth a small fraction of its Bayes-optimal ESS. Across the six models, the $c = 36$ description, which puts the mean's standard deviation at 10, is worth a median of 1.4 to 7.6 demonstrations against a Bayes-optimal 36, and at most 15 at any of $n = 1$, 2, and 4 (Figure 4a; Table 2). The $c = 1$ and $c = 4$ descriptions are worth 0.0 to 1.9 and 0.2 to 4.1, against 1.3 and 4.4. The ESS rises with c mainly because a more specific description states a value closer to the truth; as the weights show, the models make little use of the stated standard deviation. The noise-30 conditions agree (Appendix C).

The weight follows the stated specificity weakly or not at all. With no demonstrations shown and no stated reliability, the models answer with the stated value (weight 0.92 to 1.10, though Phi-4 and OLMo 2 32B then begin only 21% and 57% of their replies with a number). With one shown, the weight of gpt-oss-20b, Phi-4, and OLMo 2 32B changes by at most 0.08 from $c = 1$ to $c = 36$, and if anything falls, where the Bayes-optimal weight rises from 0.50 to 0.80 to 0.97 (Table 2). At every n from 1 to 64 and at both noise levels (Figure 4b; Table 9), the spread of these three models' weight across c is at most 0.16, against a Bayes-optimal spread of 0.34 to 0.71. The Gemma models' weight does rise with c, by about 0.25 between $c = 1$ and $c = 36$ at one demonstration (Table 2), half the Bayes-optimal rise; Qwen3.5's does not. So the models under-use a specific description and, with many demonstrations, over-use the least specific one: at 64 demonstrations gpt-oss-20b still puts weight 0.23 on the $c = 1$ value, against the Bayes-optimal 0.02, and that description raises its error (ESS -57).

Models discount a contradicted description. An irrelevant $c = 36$ description has an ESS of at most -0.4, -1.0, and -0.6 for gpt-oss-20b, Phi-4, and OLMo 2 32B, and at most -0.7 to -1.0 for the other three (bounds, because one demonstration alone beats it). By two to eight demonstrations, depending on the model, they discount a stated value that the demonstrations contradict: at $n = 4$ their weight on an irrelevant $c = 36$ value is 0.01 to 0.33, against 0.18 to 0.44 on a relevant one, so their errors (35 to 64) stay far below the 124 of the Bayes-optimal predictor, which takes it as relevant.

A stated chance of being wrong does not move the ESS in proportion. Stating a 10% or 50% chance that the description is wrong leaves the ESS of gpt-oss-20b and OLMo 2 32B about where it was, and Phi-4's even rises (from 7.6 to 8.7 and 9.7; Figure 4c). Qwen3.5 9B and Gemma 4 31B lower it, but by about as much for 50% as for 10%, and Gemma 4 12B lowers it for 10% and raises it for 50% (Table 2). No model hedges in proportion to the stated chance, while the Bayes-optimal predictor's ESS falls to 22 and 5.8. Under actual unreliability, with one description in ten irrelevant, a $c = 36$ description is worth 5.5 demonstrations to the calibrated Bayes-optimal predictor, 0.2 to the fully trusting one, and 0.4 to 3.2 to the models, or -0.1 to 4.0 when they are told the chance (Table 3).

Reasoning brings the weight close to the Bayes-optimal weight. Given room to reason before answering, gpt-oss-20b at low reasoning effort puts a weight of 0.50, 0.79, and 0.95 on the stated value with one demonstration shown, against the Bayes-optimal 0.50, 0.80, and 0.97, and comes within 0.07 of it with four (100 tasks, among the replies that finish within the token budget; Table 5). A frontier model, GPT-5.6-Luna, run inside the Codex agent, matches the Bayes-optimal weight to within 0.03, and its ESS is consistent with the Bayes-optimal ESS (32 $[25, 48]$ against 33 for the Bayes-optimal predictor on the same 100 tasks, $c = 36$; Appendix C notes the model's tool use); with its reasoning off it shows the flat weight of the open models (Tables 6 and 7).

8. Conclusion

For the Bayes-optimal predictor in in-context linear regression, a reliable description's ESS grows with the task dimension and with its specificity. With no demonstrations in hand, a description that may be irrelevant has a bounded ESS however specific it is, and once demonstrations are in hand, trusting it fully can be worse than ignoring it. Networks trained on the task distribution come close to the Bayes-optimal ESS for reliable descriptions and take their trust from training. Pretrained language models answering without reasoning adjust their weight on a description to its stated specificity far less than the Bayes-optimal predictor does, and not in proportion to its stated reliability; two models that reason first come close to the Bayes-optimal weight on a relevant description. Because the squared-error ESS needs only a learner's answers, the same question can be put to any model and to richer descriptions, including ones that are specific about some aspects of a task and silent about others.

Limitations. The analysis assumes a linear-Gaussian task and uses the robust mixture as the process that generates descriptions, so a relevant description errs only within its stated spread, an irrelevant one carries no information, and both are equally specific in every direction; a real description can be partly right, more confident than it is accurate, or specific about some aspects of a task and silent about others. The networks are trained at $d = 5$, with one or two runs per setting and no mixture output. The language-model test

covers the simplest case of the setting, six open models, and one frontier model in an agent harness; three of the open models are read from a greedy written answer, so comparisons across models mix model and readout. Its description is a stated numeric prior rather than an instruction, and other wordings change how much a model relies on it, by up to 0.8 in weight at one demonstration, without making the weight follow the stated specificity (Appendix C). The reasoning results rest on 100 tasks and, for gpt-oss-20b, on the replies that finished within the token budget. The theory and the language-model test score squared error, and the networks, which output a distribution, log loss.

Impact Statement

This paper gives task descriptions a value in demonstrations and tests whether learners rely on them accordingly, which bears on how descriptions and demonstrations are combined in prompts. We see no societal consequences to discuss.

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A. Proofs

A.1. Proof of Proposition 1 (Reliable Descriptions)

Notation and standard facts. Throughout, $\varepsilon = 1/a_0$, $X_n \in \mathbb{R}^{n\times d}$ holds the demonstration inputs as rows, $W_n = X_n^\top X_n$, and for $\kappa > 0$

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I + W_n)^{-1},$$

the error with n demonstrations under a Gaussian prior of precision κ in every direction, so that $R^{\mathrm{demo}} = R^\varepsilon$ and $R^{\mathrm{desc}}_{p=1} = R^c$. These are errors because the Bayes-optimal prediction is the posterior mean of $w^\top x$ and the query is independent of the rest of the prompt with $\mathbb{E}[xx^\top] = I$, so the error $\mathbb{E}[x^\top \Sigma x]$ is $\mathbb{E}\,\mathrm{tr}\,\Sigma$ for the posterior covariance Σ. We use three facts. (W1) For $n < d$, W_n has rank n almost surely, so $(\kappa I + W_n)^{-1}$ has the eigenvalue $1/\kappa$ with multiplicity $d - n$. (W2) For $n \geq d$ and a fixed unit vector e, $1/(e^\top W_n^{-1} e) \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12), so $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = d/(n-d-1)$ for $n \geq d+2$ and $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = \infty$ for $n \in \{d, d+1\}$. (W3) As $n \to \infty$ with d fixed, $\mathbb{E}\,\mathrm{tr}\,W_n^{-2} = d/n^2 + O(n^{-3})$ and $\mathbb{E}\,\mathrm{tr}\,W_n^{-3} = O(n^{-3})$ (von Rosen, 1988). By the Sherman–Morrison formula each new demonstration lowers $(\kappa I + W_n)^{-1}$ by a nonzero positive semidefinite matrix, so R^κ is strictly decreasing in n, and by (W2) it tends to 0.

Proof of Proposition 1. With the description and no demonstrations, $R^{\mathrm{desc}}_{p=1}(0) = d/c = ra_0 d$.

(i). By (W1), $R^\varepsilon(n) \geq (d-n)\,a_0$ for $n < d$, and trivially for $n \geq d$, so the interpolated R^{demo} satisfies $R^{\mathrm{demo}}(t) \geq (d-t)\,a_0$ at every real $t \geq 0$. For $t < (1-r)d$ this exceeds $ra_0 d$, so $\mathrm{ESS}_0 \geq (1-r)d$. As $a_0 \to \infty$, $R^\varepsilon(n)/a_0 = \mathbb{E}\,\mathrm{tr}(I + a_0 W_n)^{-1} \to (d-n)^+$ by dominated convergence, because each eigenvalue $1/(1+a_0\lambda)$ lies in $(0,1]$ and tends to 1 if $\lambda = 0$ and to 0 otherwise. So the interpolated $R^{\mathrm{demo}}(t)/a_0$ tends to $(d-t)^+$ at every t, a curve that falls through rd only at $t = (1-r)d$; as R^{demo} is decreasing, $\mathrm{ESS}_0 \to (1-r)d$.

(ii). With c fixed, $R^{\mathrm{desc}}_{p=1}(0) = d/c$ does not depend on a_0, while $R^\varepsilon(n)$ increases as a_0 grows and, by monotone convergence, (W1), and (W2), tends to $+\infty$ for $n \leq d+1$ (by (W1) for $n < d$ and (W2) for $n \in \{d, d+1\}$) and to $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = d/(n-d-1)$ for $n \geq d+2$. So the interpolated curve tends to $+\infty$ below $d+2$ and to the linear interpolation of $d/(n-d-1)$ above it, a strictly decreasing limit that, for whole c, equals d/c at $n = c + d + 1$; the root that defines ESS_0 therefore increases to $c + d + 1$. □

Extra precision with many demonstrations. Corollary 3(ii) needs the error to second order in $1/n$. Since $\kappa I + W_n$ and W_n commute,

$$(\kappa I + W_n)^{-1} = W_n^{-1} - \kappa\, W_n^{-2} + \kappa^2\, W_n^{-2}(\kappa I + W_n)^{-1},$$

and the last term lies between 0 and $\kappa^2 W_n^{-3}$. With (W2) and (W3), for fixed d and κ,

$$R^\kappa(n) = \frac{d}{n-d-1} - \frac{\kappa\, d}{n^2} + O(n^{-3}). \tag{5}$$

Each further demonstration lowers the first term by about d/n^2, and a prior of precision κ lowers the error by κ times as much, so once n is large it is worth κ demonstrations.

A.2. Proofs of Proposition 2 and Corollary 3 (Descriptions That May Be Irrelevant)

Let $D_n = (x_{1:n}, y_{1:n}, m, x)$ be the prompt with the query input, and let $\pi_n = P(Z = 1 \mid D_n)$ be the posterior probability that the description is relevant.

Step 1: the two readings. Given D_n and $Z = 1$, w is Gaussian with the posterior mean μ_1 and covariance Σ_1 of a reliable description. Given D_n and $Z = 0$, m is independent of w, and w is Gaussian with the posterior mean μ_0 and covariance Σ_0 from the demonstrations alone. At $n = 0$, $\pi_0 = p$: seeing m says nothing about Z, because m has the same law $\mathcal{N}(0, (1-r)a_0 I)$ in both cases, and x is independent of Z.

Step 2: the decomposition. The Bayes-optimal prediction is $\mathbb{E}[w^\top x \mid D_n] = (\pi_n \mu_1 + (1-\pi_n)\mu_0)^\top x$, and its error is $\mathbb{E}\,\mathrm{Var}(w^\top x \mid D_n)$. By the law of total variance over Z,

$$\mathrm{Var}(w^\top x \mid D_n) = \pi_n\, x^\top \Sigma_1 x + (1-\pi_n)\, x^\top \Sigma_0 x + \pi_n(1-\pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2.$$

For any function g of the prompt, $\mathbb{E}[\pi_n\, g(D_n)] = \mathbb{E}[Z\, g(D_n)] = p\,\mathbb{E}[g(D_n) \mid Z = 1]$, and given $Z = 1$ the prompt has the law of a prompt with a reliable description, so $\mathbb{E}[\pi_n\, x^\top \Sigma_1 x] = p\, R^{\mathrm{desc}}_{p=1}(n)$. In the same way $\mathbb{E}[(1-\pi_n)\, x^\top \Sigma_0 x] = (1-p)\, R^{\mathrm{demo}}(n)$. This is (2).

Step 3: no demonstrations. At $n = 0$, $\pi_0 = p$, $\mu_1 = m$, and $\mu_0 = 0$, so the Bayes-optimal prediction is $p\, m^\top x$. In each coordinate, $\mathrm{Var}(w_i) = a_0$, $\mathrm{Var}(m_i) = (1-r)\, a_0$, and $\mathrm{Cov}(w_i, m_i) = p\,(1-r)\, a_0$, because w_i is m_i plus independent noise when the description is relevant and is independent of m_i when it is not. Hence

$$\mathbb{E}(w_i - p\, m_i)^2 = a_0 - 2p^2(1-r)\, a_0 + p^2(1-r)\, a_0 = \big(1 - p^2(1-r)\big)\, a_0.$$

The query is independent of (w, m) with $\mathbb{E}[xx^\top] = I$, so the error $\mathbb{E}\|w - p\, m\|^2$ is d times this, which is (3).

Proof of Corollary 3. (i). At $n = 0$, (2) with $\mathcal{V}_0 \geq 0$ gives $R^{\mathrm{desc}}(0) \geq (1-p)\, R^{\mathrm{demo}}(0)$ for every r, because $R^{\mathrm{desc}}_{p=1}(0) \geq 0$. Since R^{demo} is strictly decreasing and tends to 0 (Appendix A.1), there is a finite $n_p^{\max}$ with $R^{\mathrm{demo}}(n_p^{\max}) = (1-p)\, R^{\mathrm{demo}}(0)$, and the root that defines ESS_0 cannot exceed it. By (3), $R^{\mathrm{desc}}(0) = \big(d - p^2(1-r)\, d\big)\, a_0$, and $(1-p)\, R^{\mathrm{demo}}(0) = (d - pd)\, a_0$. The bound $R^{\mathrm{demo}}(t) \geq (d-t)\, a_0$ and the limit $R^{\mathrm{demo}}(t)/a_0 \to (d-t)^+$ from the proof of Proposition 1(i) then give $\mathrm{ESS}_0 \geq p^2(1-r)\, d$, $\mathrm{ESS}_0 \to p^2(1-r)\, d$, and $n_p^{\max} \to pd$.

(ii). Fix d, a_0, and r; every O below is as $n \to \infty$. Let $L(n) = p\, R^c(n) + (1-p)\, R^\varepsilon(n)$ be the first two terms of (2). By (5), uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\mathrm{demo}}(n+t) - L(n) = \frac{d}{n^2}\Big[p\,(c - \varepsilon) - t + O(1/n)\Big].$$

The bracket is linear in t, so the same holds for the interpolated R^{demo} at real $t \in [0, T]$, and its root $n + \eta_n$, at which it equals $L(n)$, has $\eta_n = p\,(c-\varepsilon) + O(1/n)$, where $c - \varepsilon = (1-r)\, c$. If $p = 1$, then $R^{\mathrm{desc}} = L$ and $\mathrm{ESS}_n = \eta_n \to (1-r)\, c$. If $p < 1$, then $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$ and R^{demo} is decreasing, so $\mathrm{ESS}_n \leq \eta_n$ and $\limsup_n \mathrm{ESS}_n \leq p\,(1-r)\, c$. □

A.3. Derivation of Equation (4) (The Fully Trusting Predictor)

With no demonstrations the fully trusting predictor predicts $m^\top x$. With the moments of Step 3, $\mathbb{E}(w_i - m_i)^2 = a_0 - 2p(1-r)\, a_0 + (1-r)\, a_0 = \big(1 - (2p-1)(1-r)\big)\, a_0$, and summing over the d coordinates gives (4); it exceeds $R^{\mathrm{demo}}(0) = a_0 d$ exactly when $p < 1/2$. With n demonstrations it predicts $\mu_1^\top x$ whatever Z, where $\mu_1 = m + (cI + W_n)^{-1} X_n^\top (y_{1:n} - X_n m)$. As $r \to 0$ with a_0 and n fixed, $c \to \infty$ and $\mu_1 \to m$ in mean square, so its error tends to $\mathbb{E}\|w - m\|^2$: $r a_0 d \to 0$ for a relevant description and $(2-r)\, a_0 d \to 2 a_0 d$ for an irrelevant one, $2(1-p)\, a_0 d$ in all. Since $R^{\mathrm{demo}}(n) \to 0$, for every $p < 1$ there is an n from which full trust does worse than no description once r is small enough.

B. The Meta-Trained Networks in Full

Training details. Each demonstration token holds an input beside its label, and two indicator coordinates mark the token type; there is no positional encoding. The Transformer has 12 layers, 8 heads, and width 256. It is trained with Adam (learning rate 10^{-4}) for 40,000 steps at batch size 1,024, on fresh prompts at every step and without a curriculum. Without a description, the description token is masked out of attention, and the prediction is read at the query token. The labels and the description enter divided by $\sqrt{a_0}$; gradients are clipped to norm 1, and training uses bfloat16 autocast. The number of demonstrations in a training prompt is uniform on 0 to 20 (0 to 15 for the $r = 0.05$ and $r = 0.5$ networks below). The $r = 0.05$, $p = 0.9$ network of §6 reached an ESS of 3.94 after 40,000 steps and 3.93 after 80,000, against 4.98 for the Bayes-optimal predictor. With five demonstrations in hand, the two $r = 0.01$, $p = 0.9$ networks have an ESS of 7.3 and 7.7, against 8.6 for the Bayes-optimal predictor (from expected regrets on 400,000 prompts) and 8.4 for the best single Gaussian; with more demonstrations the networks' no-description curves are too flat near $n = 20$ to read the ESS reliably. Three further networks, at $r = 0.5$ ($p = 1$ and 0.9) and $r = 0.05$ ($p = 1$), reach an ESS within 0.03 of the Bayes-optimal ESS (2.25 against 2.26, 1.83 against 1.85, and 6.69 against 6.68).

C. The Language-Model Test in Full

Table 2 gives the headline values with their intervals for the six models (gpt-oss-20b under both readouts), and Tables 8 and 9 the ESS of every description at every number of demonstrations shown and the weight on the stated value, for the three models read from their distribution.

Table 1. Networks against the reference predictors ($d = 5$, $a_0 = 10$; 20,000 evaluation prompts per network, with the standard error of its ESS), grouped by training reliability; the last group is the second seed. "Trained" and "calibrated" are the Bayes-optimal predictors with the training and the test reliability, shown once where they coincide; "single Gaussian" is the best predictive that a mean and a variance can express, for a network tested at its own training reliability $p < 1$. Calibrated values, and trained values where the two reliabilities coincide, use 200,000 prompts (standard errors at most 0.12); other trained values use the network's evaluation prompts, and single-Gaussian values 100,000 prompts. "Trust" is the network's implied trust with no demonstrations, on the grid 0.001, 0.01, 0.05, 0.1, …, 0.95, 0.99, 0.999.

			ESS				
r	p_{train}	p_{test}	Network	Trained	Calibrated	Single Gaussian	Trust
0.1	1	1	5.24 ± 0.05	5.25	—	—	0.95
0.1	1	0.9	2.32 ± 0.14	2.05	4.26	—	0.95
0.01	1	1	14.74 ± 0.41	15.64	—	—	0.999
0.01	1	0.9	≤ 0	≤ 0	6.92	—	0.999
0.1	0.99	1	5.23 ± 0.05	5.26	5.25	—	0.95
0.1	0.99	0.99	5.01 ± 0.05	5.08	—	4.93	0.95
0.1	0.99	0.9	3.00 ± 0.09	3.97	4.26	—	0.95
0.01	0.99	1	11.82 ± 0.43	15.03	15.64	—	0.95
0.01	0.99	0.99	8.87 ± 0.22	12.88	—	8.54	0.95
0.01	0.99	0.9	0.07 ± 0.44	6.08	6.92	—	0.95
0.1	0.9	1	4.63 ± 0.04	5.05	5.25	—	0.85
0.1	0.9	0.9	3.57 ± 0.06	4.26	—	3.62	0.85
0.01	0.9	1	6.39 ± 0.06	11.90	15.64	—	0.85
0.01	0.9	0.9	4.30 ± 0.07	6.92	—	4.29	0.85
0.01	0.9	1	6.56 ± 0.06	11.90	15.64	—	0.90
0.01	0.9	0.99	6.27 ± 0.07	10.74	12.88	—	0.90
0.01	0.9	0.9	4.26 ± 0.08	6.92	—	4.29	0.90

Models. MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct begin only 62%, 20%, and 10% of their replies with a number, often explaining instead ("To predict…", "Given…") despite the instruction to reply with the number only, and are not reported beyond that (OLMo 2 7B's $c = 36$ ESS reads 21, from a readout resting on a minority of its probability); we report a model only where it begins at least three quarters of its replies with a number overall, and at least half of its no-description replies over $n = 1$, 2, and 4, the curve its ESS is read from. Thinking is disabled where the chat template allows it, and we start gpt-oss-20b's reply in its final channel. The models run in Hugging Face Transformers at their native precision (`openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, `google/gemma-4-31B-it`), and tasks come from a seeded generator.

Table 2. The ESS and the weight on the stated value for each model. Left: the ESS of a description (median over $n = 1$, 2, and 4 demonstrations shown) for a relevant description with $c = 1$, 4, or 36, and for the $c = 36$ description when irrelevant or with a stated 10% or 50% chance of being wrong. Brackets, 95% bootstrap intervals; $\leq$, an upper bound, where one demonstration alone beats the description; $\sigma = 60$, 2,000 tasks; "written": read from the number the model writes. Right: the weight on the stated value with one demonstration shown; the Bayes-optimal weight is $c/(c+n)$. The Bayes-optimal predictor takes a description as relevant unless a chance of being wrong is stated, and then hedges with that chance.

	ESS, median over $n = 1, 2, 4$						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	stated 10%	stated 50%	$c = 1$	$c = 4$	$c = 36$
Bayes-optimal	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

Prompt. The user turn of the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.
548
431
Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." With a stated reliability, a third sentence follows: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." With no demonstrations, the question is "Predict the first number." The stated value of a relevant description is drawn given the task, so that $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$; a task is redrawn in the rare case (under 1%) that a number or stated value falls outside 100 to 999.

Measurement. The ESS of §7 is Definition 1 with squared error. Its summary leaves out $n = 0$, where without a description the models rarely answer with a number, and large n, where the ESS mixes the description's value with the models' slow learning (error 9 to 23 after 64 demonstrations at $\sigma = 60$, against 7.5 for the Bayes-optimal predictor) and their over-reliance on loose descriptions. A model that learns slowly from demonstrations values a description more in its own units; the weights do not depend on this. The no-description mean squared error at $n = 1, 2, 4, \ldots, 64$ is made nonincreasing by a least-squares monotone fit (pool adjacent violators) and interpolated linearly in $\log n$, on the squared error, not its root; ESS_n is the t at which that curve equals the description's mean squared error at n, minus n. Since the curve starts at one demonstration, when one demonstration alone beats the description we report the bound $\mathrm{ESS}_n \leq 1 - n$. Each bootstrap replicate resamples tasks, the same tasks for both curves, and repeats the fit and the interpolation, so the intervals reflect sampling over tasks under this procedure, not the interpolation bias quantified next.

The Bayes-optimal reference. The Bayes-optimal predictor is that of §7, with the mean uniform over 100 to 999, so that without a description its prediction is essentially the mean of the demonstrations shown (the range barely truncates it). A relevant description's exact Bayes-optimal ESS is then c, up to that truncation, that is 1, 4, and 36. Computed the same way as the models' ESS, it reads 1.3, 4.4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$: the $\log n$ interpolation biases it upward (by about 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36 with exact errors), and the finite sample of tasks shifts it further, in either direction (for $c = 36$, down to 36 at $\sigma = 60$ and up to 43 at $\sigma = 30$). The tables report the computed value for the Bayes-optimal predictor and the models alike. Under the prior the tasks are drawn from, $\mathcal{N}(500, 100^2)$, a predictor without the description would hold a prior precision of $\sigma^2/100^2 = 0.36$ demonstrations at $\sigma = 60$ (0.09 at $\sigma = 30$), and a relevant description's exact ESS would be $c - 0.36$, that is 0.64, 3.64, and 35.64, the $(1 - r)\,c$ of §4. We keep the predictor that knows only what the prompt states, since the models are told nothing else.

Replies. A model's share of replies that begin with a number is the probability its first token puts on the number tokens; where that share is below three quarters, Table 9 sets the cell in italics. The noise-30 conditions, which realize the same three c with both standard deviations in the prompt halved, show the same pattern as the noise-60 ones (Table 8). OLMo 2 32B's first token puts 41% of its probability on numbers in the no-description prompts with one demonstration shown and 75% with two, so its no-description prediction at $n = 1$ rests on a minority of that probability.

Written readout. For Qwen3.5 9B, Gemma 4 12B, and Gemma 4 31B, and as a check on the others, the prediction is the first three-digit integer in the model's reply under greedy decoding with eight new tokens; a reply without one is dropped from the error curves that the ESS is read from, and in the weight regression it is replaced by the average of the model's answers in that condition; the share of replies with one is the "number" column. On gpt-oss-20b the written readout gives weights within 0.2 of the distribution readout's in every condition and at every n, and headline ESS medians of 0.2, 2.0, and 3.7 against 0.9, 2.0, and 2.8; it is the noisier readout, since one greedy answer stands in for the distribution's mean. Phi-4 begins only 63% of its written replies with a number over all conditions, so Table 2 reads it from its distribution only; on the wording ablation's grid its share is above three quarters for the paper's wording. OLMo 2 32B writes a number in only 5%, 36%, and 48% of its no-description replies at $n = 1$, 2, and 4 ($\sigma = 60$), so its written ESS is not reported; its written weights on the three relevant descriptions at $n = 1$ are 0.70, 0.74, and 0.75, against 0.72, 0.70, and 0.70 from its distribution.

Actual unreliability. The stated-reliability conditions keep the description relevant and test whether a model hedges as told. Table 3 instead realizes the mixture of §3: the squared errors of the relevant and the irrelevant condition, on the same

Table 3. **ESS under actual unreliability.** The relevant and irrelevant conditions combined with weights p and $1-p$; median over $n = 1$, 2, and 4 demonstrations shown, with 95% bootstrap intervals over tasks; $\sigma = 60$, 2,000 tasks. A bound where one demonstration alone beats the description at some n.

	irrelevant one time in ten ($p = 0.9$)		irrelevant half the time ($p = 0.5$)	
Predictor	$c = 4$	$c = 36$	$c = 4$	$c = 36$
Bayes-optimal, takes it as relevant	1.0 [0.8, 1.1]	0.2 [0.0, 0.4]	≤ -1.0	≤ -1.0
Bayes-optimal, hedges at p	2.3 [2.1, 2.7]	5.5 [5.1, 6.0]	0.6 [0.5, 0.7]	0.9 [0.8, 1.1]
gpt-oss-20b, not told	1.2 [1.0, 1.4]	1.6 [1.4, 1.7]	0.1 [0.0, 0.2]	≤ 0.0
gpt-oss-20b, told	1.1 [1.0, 1.3]	1.3 [1.1, 1.4]	-0.1 [-0.1, 0.0]	-0.1 [-0.1, 0.0]
Phi-4, not told	1.3 [1.1, 1.5]	1.9 [1.7, 2.4]	≤ -1.0	≤ -1.0
Phi-4, told	2.7 [2.2, 3.2]	3.6 [3.1, 4.1]	≤ 0.0	≤ 0.0
OLMo 2 32B, not told	2.5 [2.2, 2.8]	3.2 [2.9, 3.5]	≤ 0.7	≤ 0.6
OLMo 2 32B, told	3.2 [2.9, 3.4]	4.0 [3.6, 4.3]	≤ 1.0	≤ 0.9
Qwen3.5 9B (written), not told	1.0 [0.7, 1.2]	1.8 [1.3, 2.1]	≤ -0.3	≤ -0.9
Qwen3.5 9B (written), told	-1.0 [-1.0, -0.7]	-0.1 [-0.5, 0.3]	≤ -1.0	≤ -1.0
Gemma 4 12B (written), not told	0.3 [0.2, 0.5]	0.7 [0.5, 1.0]	≤ -0.1	≤ 0.0
Gemma 4 12B (written), told	0.6 [0.5, 0.7]	1.0 [0.9, 1.3]	≤ 0.0	≤ 0.1
Gemma 4 31B (written), not told	0.1 [0.0, 0.2]	0.4 [0.3, 0.5]	≤ 0.0	≤ -0.1
Gemma 4 31B (written), told	≤ 0.0	0.2 [0.1, 0.3]	≤ 0.0	≤ 0.0

tasks, are combined with weights p and $1-p$ before the ESS is read, for a model not told the chance and for one told it, against the fully trusting Bayes-optimal predictor and the calibrated one, which hedges at p. With one description in ten irrelevant, the $c = 36$ description is worth 5.5 demonstrations to the calibrated predictor and 0.2 to the fully trusting one; to the models it is worth 0.4 to 3.2 when not told and -0.1 to 4.0 when told. Their flat weight on a description acts as a hedge between full trust and calibration, and being told the chance moves it by at most two demonstrations, in either direction. With half the descriptions irrelevant, the calibrated predictor keeps under one demonstration and no model keeps more than one.

Wording of the description. We reran the three relevant descriptions and the no-description prompt at $n = 1$, 2, and 4 with three other wordings of the description sentence: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." (Table 4; written readout, 2,000 tasks). The wording sets how much a model relies on the description: with one demonstration shown, Gemma 4 12B's weight on the $c = 1$ value is 0.34 under the paper's wording and 1.00 under the other three, and gpt-oss-20b's ranges from 0.17 to 0.67 (a weight above 1 means the prediction moves more than the stated value does). Under no wording does the weight follow the stated specificity as the Bayes-optimal weight does: the gap between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$ over all models and wordings, against 0.47 and 0.70 for the Bayes-optimal predictor.

Reasoning. gpt-oss-20b was also run with its reasoning on at low effort, on 100 tasks drawn by the same generator, with 384 new tokens and the reply read from its final channel, for seven conditions at $n = 1$ and 4 (Table 5). When a reliability is stated it answers with a number in only 28% to 54% of replies, the budget running out before the answer; those cells rest on the replies that finished and are set in italics. Where it answers, its weight on the stated value is the Bayes-optimal weight to within 0.02 for a relevant description at $n = 1$ and within 0.07 at $n = 4$, it discounts an irrelevant one a little (0.91 against 0.97 for the predictor that takes it as relevant), and at $n = 1$ it separates a 10% chance of being wrong from a 50% one (0.96 and 0.70, from a third to a half of its replies, where the Bayes-optimal hedge on the same tasks is 0.90 and 0.49), by more than any model without reasoning; at $n = 4$ the order reverses (0.74 and 0.77, against 0.79 and 0.70).

A frontier model inside an agent harness. The models above are open ones run locally, so that their first-token distributions can be read. The squared-error ESS needs only answers, so we also put the prompts of three relevant descriptions ($c = 1$, 4, 36; $\sigma = 60$) with $n = 1$ and 4 demonstrations, 100 tasks each, to GPT-5.6-Luna (model `gpt-5.6-luna`) through the Codex command-line agent in its read-only sandbox, on a subscription, once with its reasoning effort set to low (about 115 reasoning tokens per answer) and once with reasoning off. This measures the model as deployed in that agent: its instructions and tool definitions, about 26,000 tokens, surround the prompt, as a provider's system prompt surrounds it through an API. The model used no tool in any of these 1,200 replies, and every reply is a number alone (ten with a trailing period; 181 with a decimal part, which we keep).

Table 4. **The wording of the description sets how much a model relies on it, but under no wording does the weight follow the stated specificity.** Weight on the stated value for the three relevant descriptions at $n = 1$ and 4 under four wordings (written readout, 2,000 tasks). Number: the share of replies with a number; italics: below three quarters in that cell. The Bayes-optimal weight is 0.50, 0.80, and 0.97 at $n = 1$ and 0.20, 0.50, and 0.90 at $n = 4$.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (the paper's)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (the paper's)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (the paper's)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (the paper's)	0.61	0.70	0.74	0.75	*0.19*	*0.27*	0.32
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.45	0.47	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.93	0.95	0.50	0.57	0.64
Phi-4	normal, mean m, sd τ (the paper's)	0.77	0.88	0.58	0.63	*0.27*	*0.15*	*0.24*
	about m, give or take τ	0.97	0.99	0.94	0.96	0.65	0.59	0.63
	Mean: $m \pm \tau$	0.83	*0.73*	*0.58*	*0.66*	0.51	0.49	0.56
	earlier estimate m, s.e. τ	0.81	0.78	0.71	0.71	*0.48*	*0.42*	0.53
Qwen3.5 9B	normal, mean m, sd τ (the paper's)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.37
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

Table 5. **With reasoning, gpt-oss-20b puts close to the Bayes-optimal weight on a relevant description.** Low reasoning effort, 100 tasks, $\sigma = 60$. Number: the fraction of replies that are a number. Weight and error: the model's against the Bayes-optimal predictor's on the tasks where the model gave a number; the predictor hedges with the stated p where one is given. Italics: fewer than three quarters of the replies are a number.

	$n = 1$			$n = 4$		
Description	number	weight	error	number	weight	error
no description	1.00	–	60/60	1.00	–	30/30
rel. $\tau = 60$	1.00	0.50/0.50	45/45	0.80	0.13/0.20	34/29
rel. $\tau = 30$	0.96	0.79/0.80	28/27	0.71	*0.52/0.50*	*23/19*
rel. $\tau = 10$	0.93	0.95/0.97	13/9	0.87	0.88/0.90	10/9
irrel. $\tau = 10$	0.94	0.91/0.97	119/123	0.78	0.84/0.90	113/113
rel. $\tau = 10$, 10% w.	0.54	*0.96/0.90*	*11/11*	0.43	*0.74/0.79*	*22/11*
rel. $\tau = 10$, 50% w.	0.33	*0.70/0.49*	*28/36*	0.28	*0.77/0.70*	*18/11*

With reasoning, the weight on the stated value is the Bayes-optimal $c/(c+n)$ to within 0.03 in all six cells, the implied worth $n\,a/b$, with a and b the weights on the stated value and on the demonstrations' mean, is c to within half a demonstration (Table 6), and the error matches the Bayes-optimal predictor's to within 0.8. The prompt therefore contains what is needed to reach the Bayes-optimal weight, and this model reaches it when it reasons. With reasoning off, the same model shows the flat weight of §7: 0.72 to 0.88 at $n = 1$ and 0.40 to 0.47 at $n = 4$ whatever the stated specificity, and an implied worth of 2.6 to 5.8 demonstrations for descriptions worth 1 to 36.

To read the ESS of Definition 1 from this model's own no-description curve, we also sent the no-description prompts on the same 100 tasks at $n = 1, 2, 4, \ldots, 64$ in both modes (1,400 replies, all numbers). With reasoning, the model averaged the

shown numbers with a shell command in the agent's sandbox (Python in 127 prompts, Node.js or awk in the rest) in 174 of these 700 prompts (4 at $n = 16$, 74 at $n = 32$, and 96 at $n = 64$), and its no-description error matches the Bayes-optimal predictor's within 0.1 at every n (for example 9.4 and 7.0 at $n = 32$ and 64, against 9.4 and 6.9); with reasoning off it used no tool. A more accurate no-description curve lowers the ESS read from it, so the tool use does not inflate the ESS. With reasoning, the ESS of the three descriptions at $n = 1$ and 4 matches the Bayes-optimal predictor's, read the same way from its predictions on the same tasks, to within 0.2 demonstrations for $c = 1$ and 4 and 32 against 33 for $c = 36$ (Table 7; with 100 tasks the reading itself moves the Bayes-optimal 4 to 2.4 and 2.9). With reasoning off, the $c = 36$ description is worth 5.4 demonstrations at $n = 1$ and 1.3 at $n = 4$, in the range of the open models of §7, and the $c = 1$ description hurts at $n = 4$.

Table 6. **A frontier model reaches the Bayes-optimal weight when it reasons and shows the flat weight of the open models when it does not.** GPT-5.6-Luna inside the Codex agent harness, 100 tasks per cell, relevant descriptions at $\sigma = 60$; weight on the stated value and implied worth $n\,a/b$ from the regression of §7; bootstrap standard errors of the weights are at most 0.03 with reasoning and 0.07 without.

	weight on the stated value						implied worth, demonstrations					
	$n = 1$			$n = 4$			$n = 1$			$n = 4$		
Bayes-optimal ESS c	1	4	36	1	4	36	1	4	36	1	4	36
Bayes-optimal	0.50	0.80	0.97	0.20	0.50	0.90	1	4	36	1	4	36
GPT-5.6-Luna, low reasoning	0.50	0.77	0.97	0.21	0.50	0.90	1.0	3.6	35.7	1.0	4.0	36.5
GPT-5.6-Luna, reasoning off	0.79	0.72	0.88	0.43	0.40	0.47	3.6	2.6	5.8	3.1	2.7	3.7

Table 7. **When it reasons, GPT-5.6-Luna's ESS is consistent with the Bayes-optimal ESS.** ESS at $n = 1$ and 4 demonstrations shown, read from the model's own no-description curve ($n = 1$ to 64, the same 100 tasks; with reasoning, the model averaged the numbers with a shell command in 174 of these 700 prompts, mostly at $n = 32$ and 64) as in §7, with 95% bootstrap intervals over tasks; the Bayes-optimal predictor's predictions on the same tasks are read the same way.

	$n = 1$			$n = 4$		
Bayes-optimal ESS c	1	4	36	1	4	36
Bayes-optimal, read the same way	1.0 [0.6, 1.8]	2.4 [1.6, 3.9]	33 [25, 49]	1.7 [0.5, 3.6]	2.9 [1.2, 7.7]	33 [23, 50]
GPT-5.6-Luna, low reasoning	1.1 [0.6, 1.8]	2.3 [1.4, 3.8]	32 [25, 48]	1.5 [0.3, 3.4]	3.0 [1.2, 7.7]	32 [23, 49]
GPT-5.6-Luna, reasoning off	0.1 [0.0, 0.5]	1.2 [0.8, 2.1]	5.4 [2.5, 14]	−2.2 [−2.6, −1.4]	0.1 [−0.7, 2.5]	1.3 [−0.2, 3.8]

Table 8. ESS of a description for every condition and number of demonstrations shown, for the Bayes-optimal predictor and the three models read from their distribution (2,000 tasks). "number" is the share of a model's replies that begin with a three-digit number, averaged over n; "10% wrong" and "50% wrong" mark a stated chance of being wrong. The Bayes-optimal predictor is that of §7, optimal for what the prompt states: it takes an irrelevant description with no stated chance as relevant, so its ESS can be negative. $\leq$, an upper bound; $> k$, the error with the description is below the no-description error at 64 demonstrations, so the ESS exceeds $k = 64 - n$.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Bayes-optimal	$\sigma = 60$, $c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60$, $c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60$, $c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60$, $c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60$, $c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30$, $c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30$, $c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30$, $c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60$, $c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60$, $c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60$, $c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60$, $c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60$, $c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60$, $c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60$, $c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60$, $c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60$, $c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30$, $c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30$, $c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30$, $c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60$, $c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-2.9	-34
	$\sigma = 60$, $c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.7	-40
	$\sigma = 60$, $c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.4	-33
	$\sigma = 60$, $c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60$, $c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.6	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30$, $c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30$, $c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30$, $c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	16	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.85	≤ 0.0	≤ -1.0	-2.1	-1.2	2.9	-3.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.86	≤ 0.0	≤ -1.0	-2.4	-1.7	1.3	-4.5	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.85	≤ 0.0	≤ -1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	≤ 0.0	≤ -1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60$, $c = 1$, irrelevant	0.89	≤ 0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60$, $c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant	0.90	≤ 0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60$, $c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.90	≤ 0.0	≤ -1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30$, $c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30$, $c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30$, $c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	≤ -1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table 9. Weight on the stated value for every condition and number of demonstrations shown, for the Bayes-optimal predictor and the three models read from their distribution; "number" as in Table 8. A cell is in italics where the model begins fewer than three quarters of its replies with a number at that n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Bayes-optimal	$\sigma=60, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60, c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60, c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60, c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60, c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60, c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60, c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60, c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60, c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60, c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60, c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60, c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30, c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30, c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30, c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60, c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60, c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60, c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60, c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60, c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60, c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60, c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60, c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60, c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30, c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30, c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30, c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60, c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma=60, c=1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma=60, c=1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma=60, c=4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma=60, c=4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma=60, c=36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma=60, c=36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma=30, c=1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma=30, c=4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma=30, c=36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma=60, c=36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma=60, c=4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma=60, c=36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01